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SARDAR PATEL UNIVERSITY
BCA Semester-5 External Examination – 2026

[NEP]

US05MABCA01 || Computer Networks

Date : 09-12-2026

Time : 10.00 a.m. – 12.30 p.m.

Total : 50 Marks



Q-1 Select the appropriate option for the following questions:

[10]

- 1) What does a network help improve in communication?
(A) File size (B) Signal strength
(C) Information sharing (D) Battery life
- 2) What is it called when a packet is sent to all machines in a network?
(A) Unicasting (B) Unicasting
(C) Broadcasting (D) Narrowcasting
- 3) A _____ cable is made of glass or plastic and transmits signals in the form of light.
(A) Twisted-pair (B) Coaxial cables
(C) Fiber Optics (D) None of these
- 4) _____ multiplexing method is used for digital signals.
(A) WDM (B) FDM
(C) TDM (D) CDMA
- 5) A _____ is a set of rules that govern data communications.
(A) Layers (B) Network
(C) Protocol (D) Communication
- 6) Routing is responsibility of _____.
(A) Presentation layer (B) Network layer
(C) Physical layer (D) Session layer
- 7) Which of the following topology requires maximum cabling?
(A) Bus (B) Star
(C) Ring (D) Mesh
- 8) A repeater operates at layer _____.
(A) Physical (B) Data link
(C) Network (D) All of above
- 9) What is a star, bus, or ring in networking?
(A) Protocol (B) Topology
(C) Server (D) Host
- 10) _____ Channels are required in serial transmission.
(A) Multiple (B) One
(C) Two (D) Four

Q-2 Explain the term Computer Network along with its advantages and disadvantages.

[10]

OR

Q-2 Explain Local Area Network (LAN) in detail. Mention its topology, speed, error rate, and commonly used protocols. [10]

Q-3 Describe the construction, types, and uses of twisted pair cables. Explain the differences between Shielded Twisted Pair and Unshielded Twisted Pair cables. [10]

[1]

(P.T.O.)

OR

Q-3 Differentiate between Parallel and Serial Transmission. Explain with diagrams and mention their respective advantages and limitations. **[10]**

Q-4 What is OSI Model? Explain any three layers of OSI Model in detail along with diagram. **[10]**

OR

Q-4 Write a note on satellite Network. **[10]**

Q-5 Explain Bus, Ring and Star topologies along with its advantages and disadvantages. **[10]**

OR

Q-5 Describe Modem, Repeater, Router and Bridge in detail. **[10]**

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